ABDULRAHMAN ALSAADI

MECHATRONICS & ROBOTICS ENGINEER

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PROFESSIONAL SUMMARY

Mechatronics and Robotics Engineer focused on robotics, autonomous systems, and R&D, with hands-on experience integrating robotics software, embedded control, computer vision, and mechanical design. Experienced in taking robotic systems from simulation and engineering analysis through hardware integration, testing, and physical validation.

TECHNICAL SKILLS

Robotics & Autonomy: ROS 2, Gazebo, RViz, URDF/Xacro, ArduPilot SITL, MAVLink, pymavlink, Forward & Inverse Kinematics
Embedded Systems & Control: ESP32, Raspberry Pi 5, Arduino, Cube Orange, UART, PWM, MQTT, Stepper, Servo & BLDC Motors, Systems Integration
Programming & Computer Vision: C++, Python, OpenCV, Linux (Ubuntu)
Mechanical Design, Analysis & Validation: SolidWorks, SolidWorks Simulation, FEA, Topology Optimization, DFM, Dimensional Tolerancing, Fits & Clearances, Bearing Fits, BOM Development, 3D Printing, Manufacturing Drawings

SELECTED ENGINEERING PROJECTS

WD DRONE - TEKNOFEST 2026, Rotary Wing Category — Karabük, Türkiye

Autonomous Control & Simulation — **Feb 2026 - Jul 2026**

- Built and validated a custom **Gazebo + ArduPilot SITL** mission environment for autonomous waypoint search, visual target detection and approach, payload release, and return-to-launch; transferred the mission architecture to the Raspberry Pi 5 and Cube Orange platform.
- Integrated **Raspberry Pi 5, Camera Module 3, and Cube Orange** through MAVLink over UART using pymavlink; implemented OpenCV color and geometric target detection at approximately **30 FPS**, flight-mode switching, and payload-release control.
- Validated the UAV through staged hardware and flight testing, including an **8 kg structural load test, 12-minute unloaded hover, 7-minute hover with a 1 kg payload, and five flight-test sessions**; supported troubleshooting of severe flight oscillation through PID gain adjustment.
- Contributed to system requirements, UAV architecture, component selection, BOM development, technical documentation, and coordination across the six-member multidisciplinary team.

SCARA ROBOTIC ARM - GRADUATION PROJECT — Karabük, Türkiye

Mechatronics & Robotics Engineer — **Apr 2026 - Jun 2026**

- Designed, manufactured, assembled, and physically validated a custom **3-DOF SCARA robotic arm**, achieving a verified **492.5 mm reach and 100 g payload at full extension**; applied DFM, dimensional tolerancing, bearing-fit principles, BOM development, and manufacturing drawings.
- Reduced **Arm Link 1 mass by approximately 16.5%** through SolidWorks FEA and topology optimization while retaining critical load-bearing regions; designed a two-stage GT2 belt transmission, completed joint torque calculations, and manufactured functional PLA+ components.
- Implemented ESP32/C++ control with automatic homing, forward and inverse kinematics, Cartesian positioning, programmable motion sequences, and servo-gripper control; integrated the robot through **MQTT** with a vision-enabled conveyor and mobile robot for automated box transfer.

EXPERIENCE

SMART METHODS — Jeddah, Saudi Arabia

Full Stack Robotics Engineering Intern — **Jul 2025 - Aug 2025**

- Designed an early-stage **5-DOF robotic-arm concept in SolidWorks**, including joint redesign, static analysis, torque calculations, servo selection, and preparation of components for additive manufacturing.
- Assembled and programmed a physical 5-DOF robotic-arm kit and developed web/mobile control interfaces for storing, retrieving, and replaying robot poses using **ESP32, PHP/MySQL, and Flutter**.
- Completed robotics exercises covering embedded control, forward kinematics, computer vision, sensing, motor actuation, and hardware testing using Arduino, ESP32, Python, and OpenCV

WASH THRU — Jeddah, Saudi Arabia

Industrial Automation Panel Assembly Intern — **Jun 2025 - Jul 2025**

- Assembled customized industrial control panels from electrical schematics, working with **PLCs, VFDs, contactors, relays, motor protection devices, energy meters, and terminal components**.
- Performed wire preparation, crimping, termination, component placement, and cable management while documenting assembly activities for technical reporting.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering — **Jun 2021 - Jun 2026**
Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- Certified SOLIDWORKS Professional - Mechanical Design (CSWP), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate - Mechanical Design (CSWA), Dassault Systèmes | May 2026
- Saudi Council of Engineers (SCE) - Engineer Membership (Active)
- ROS 2 Fundamentals, Karabük University | May 2025

LANGUAGES

Arabic - Native | English - Upper-intermediate | Turkish - Intermediate